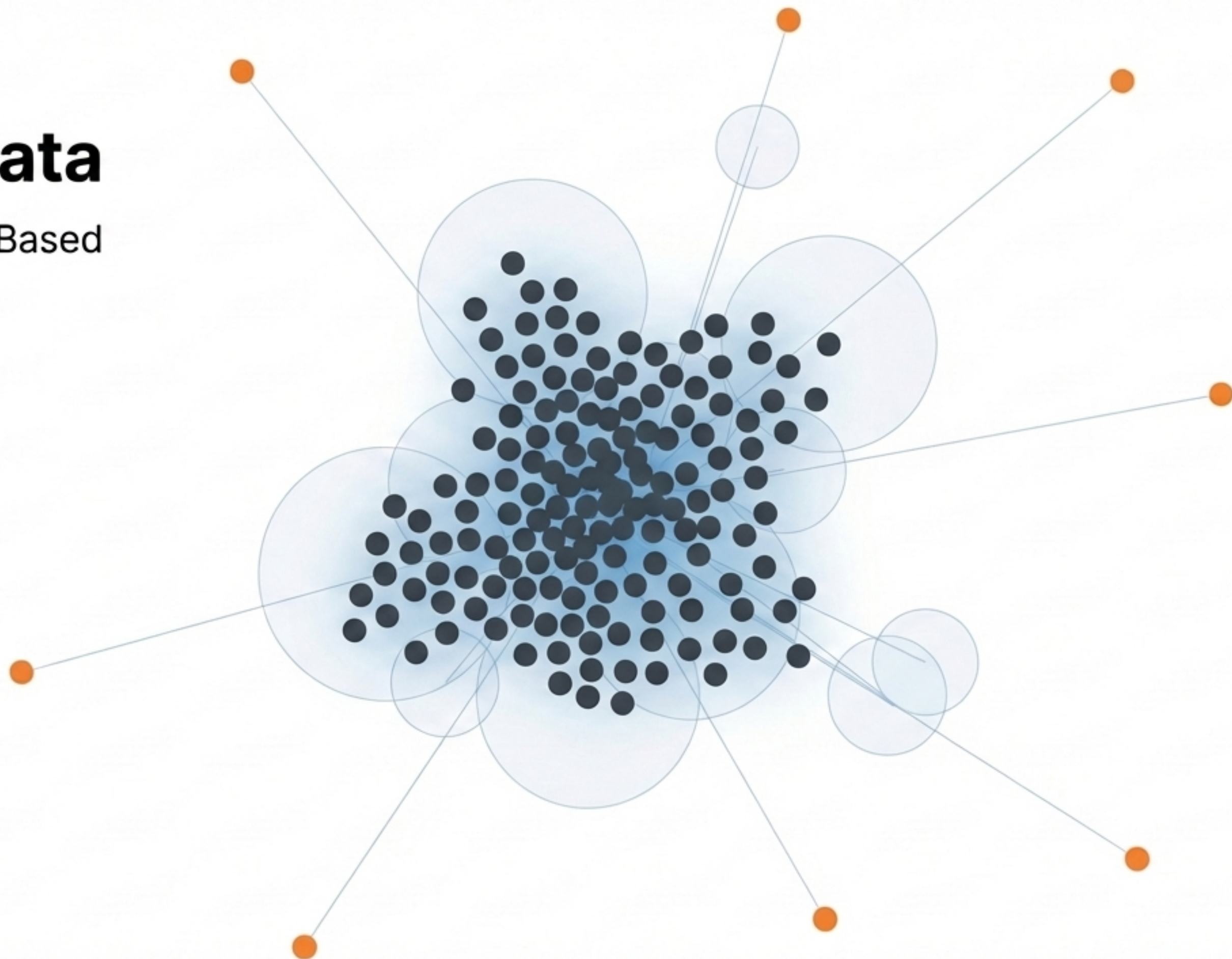
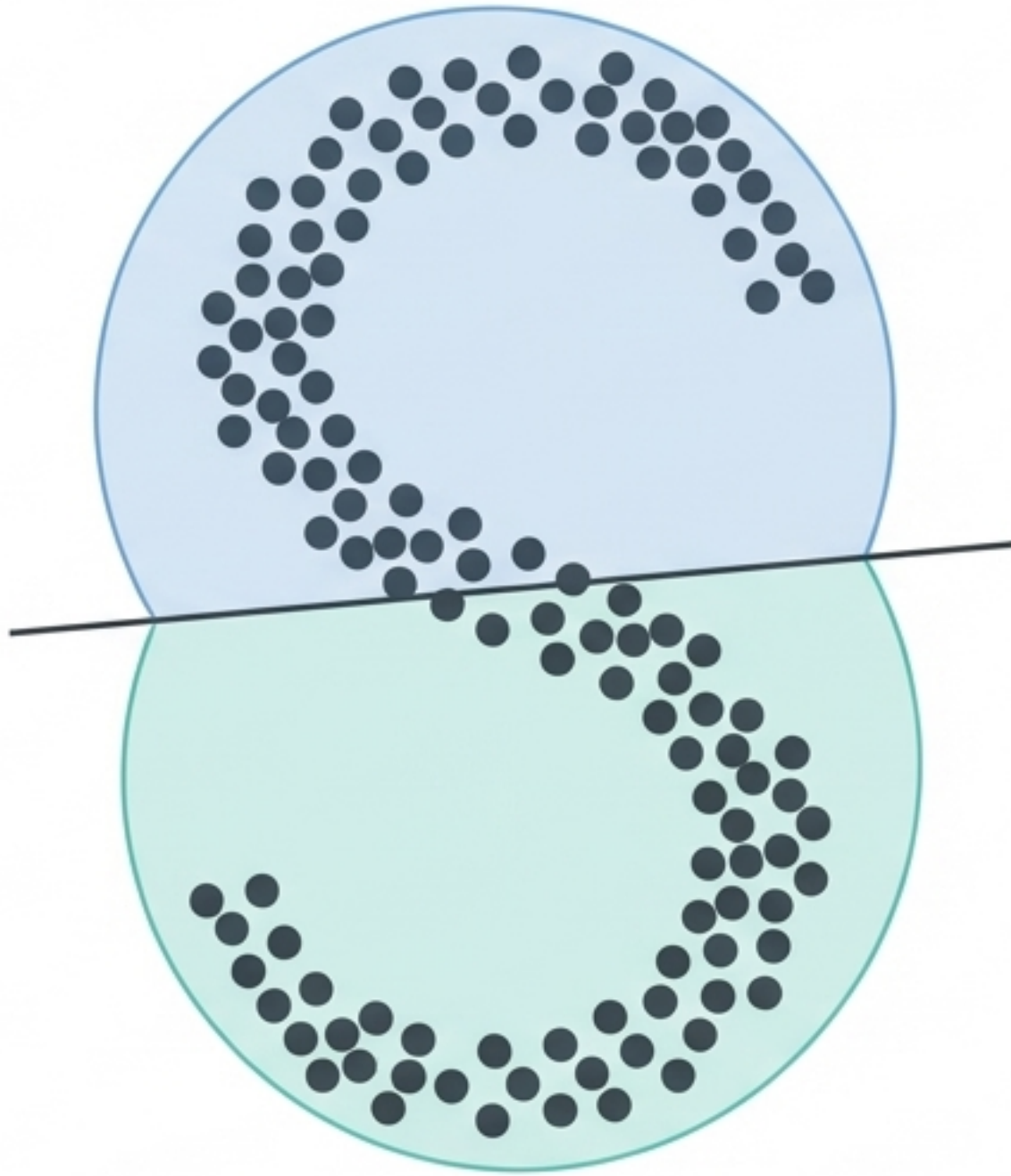


Mapping the Geometry of Data

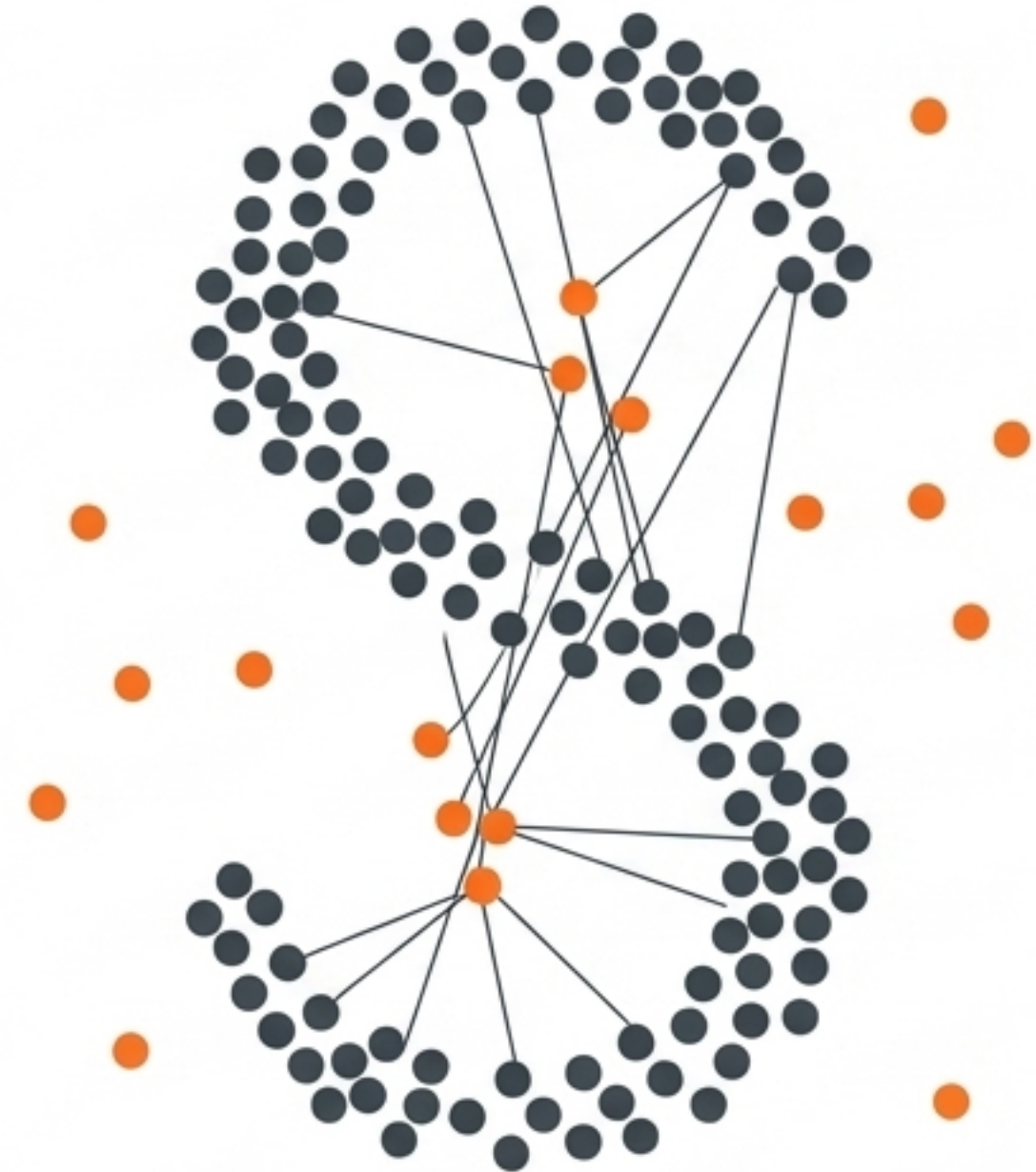
An Introduction to Density-Based Clustering and DBSCAN



The spherical trap of traditional clustering methods

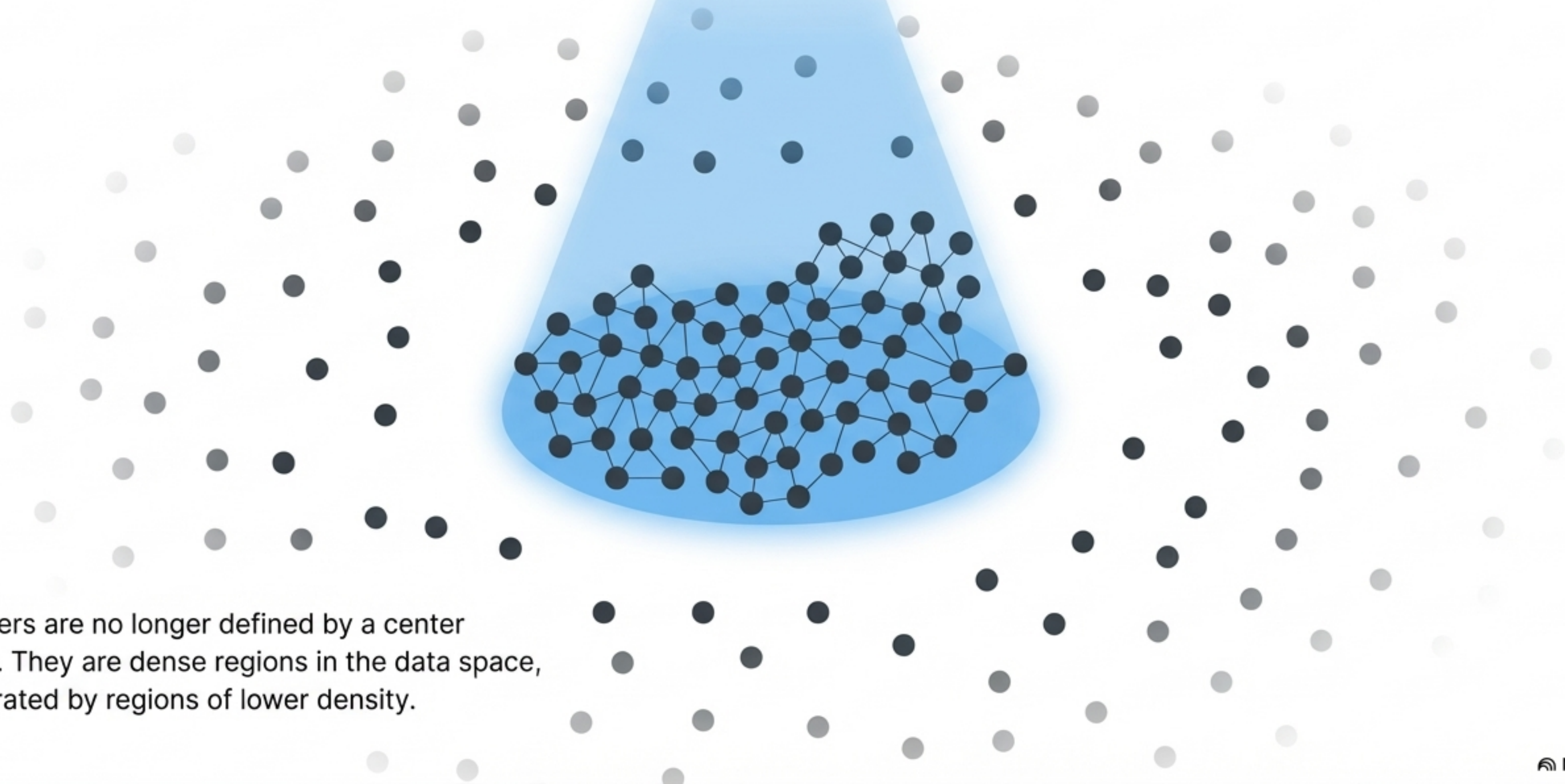


→ Forces spherical partitions



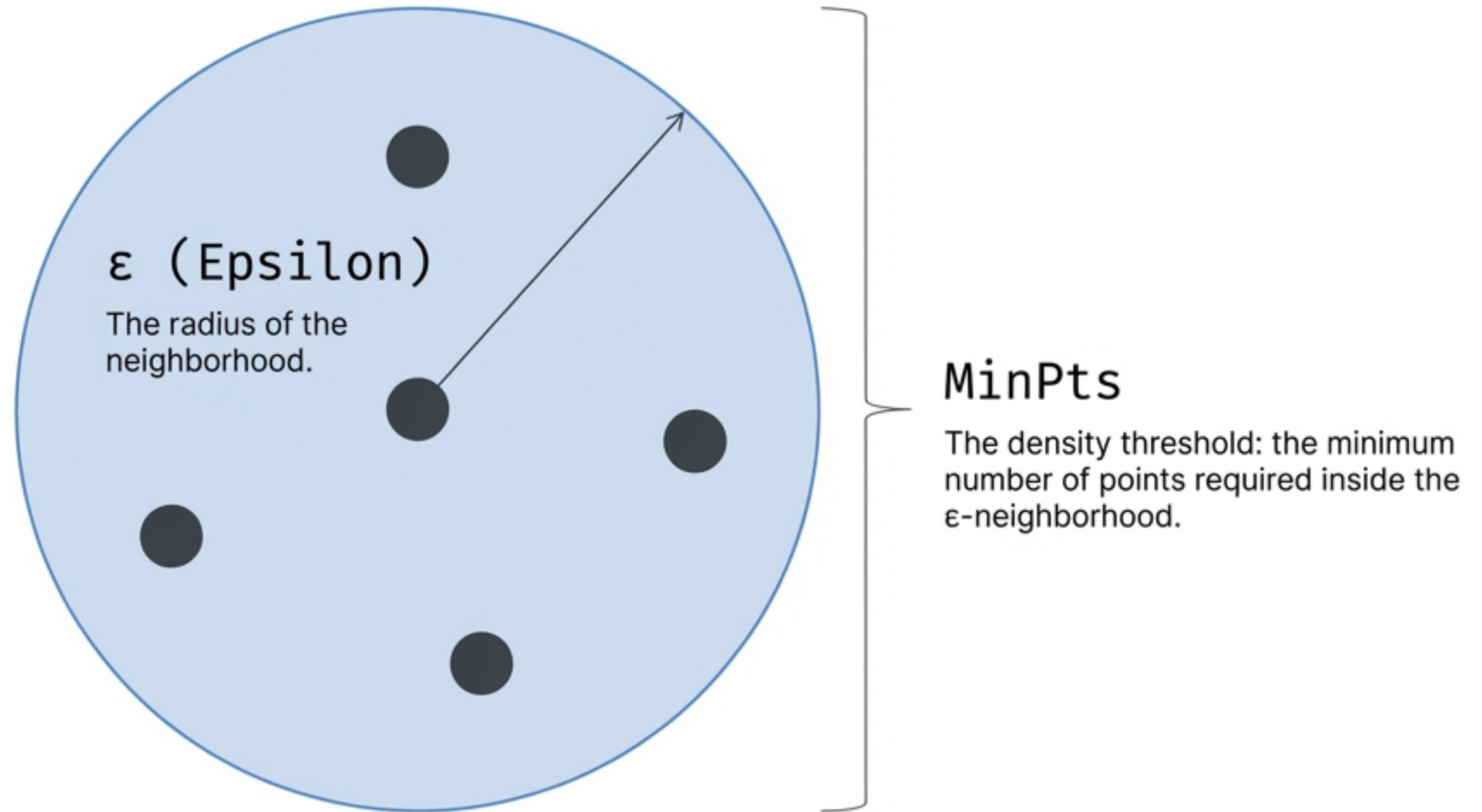
→ Highly sensitive to noise bridges

Shifting the paradigm from distance to density

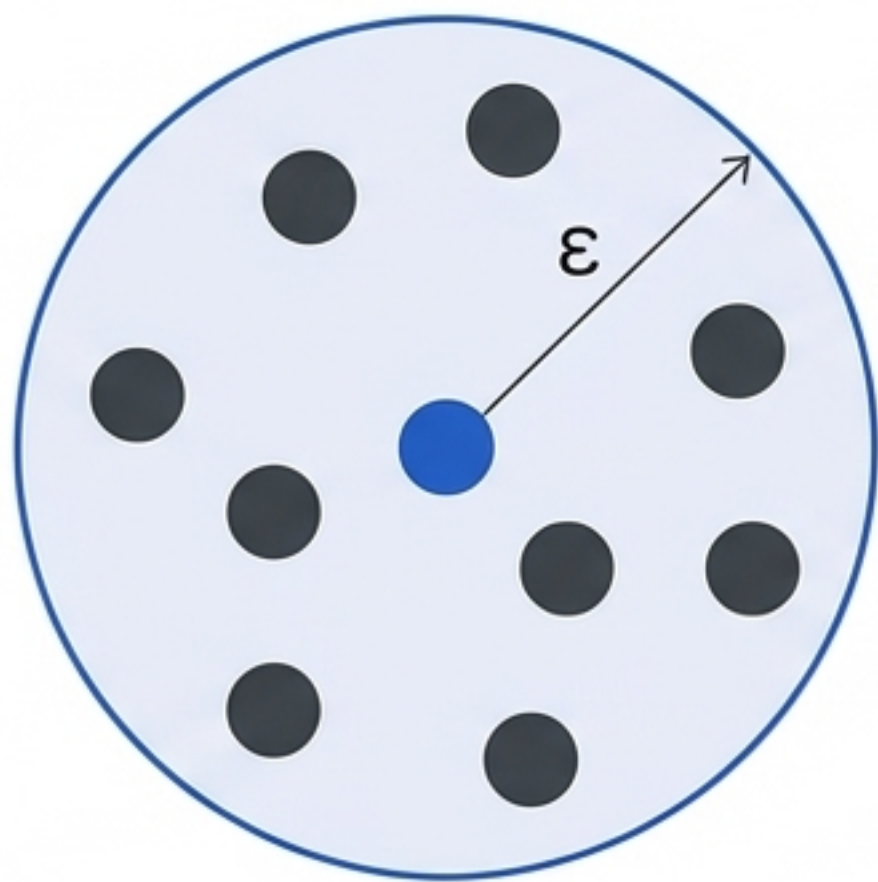


Clusters are no longer defined by a center point. They are dense regions in the data space, separated by regions of lower density.

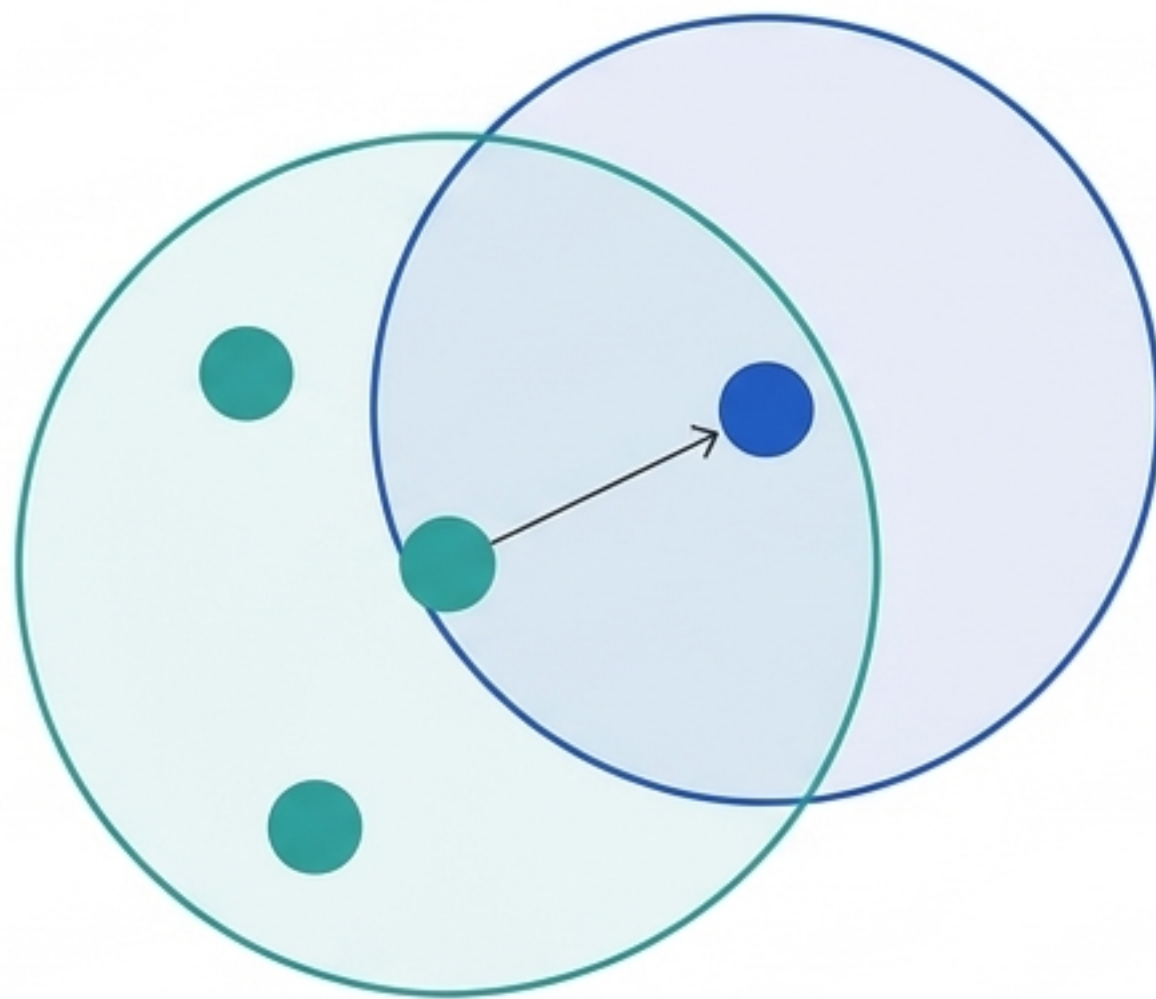
The two geometric rules of density



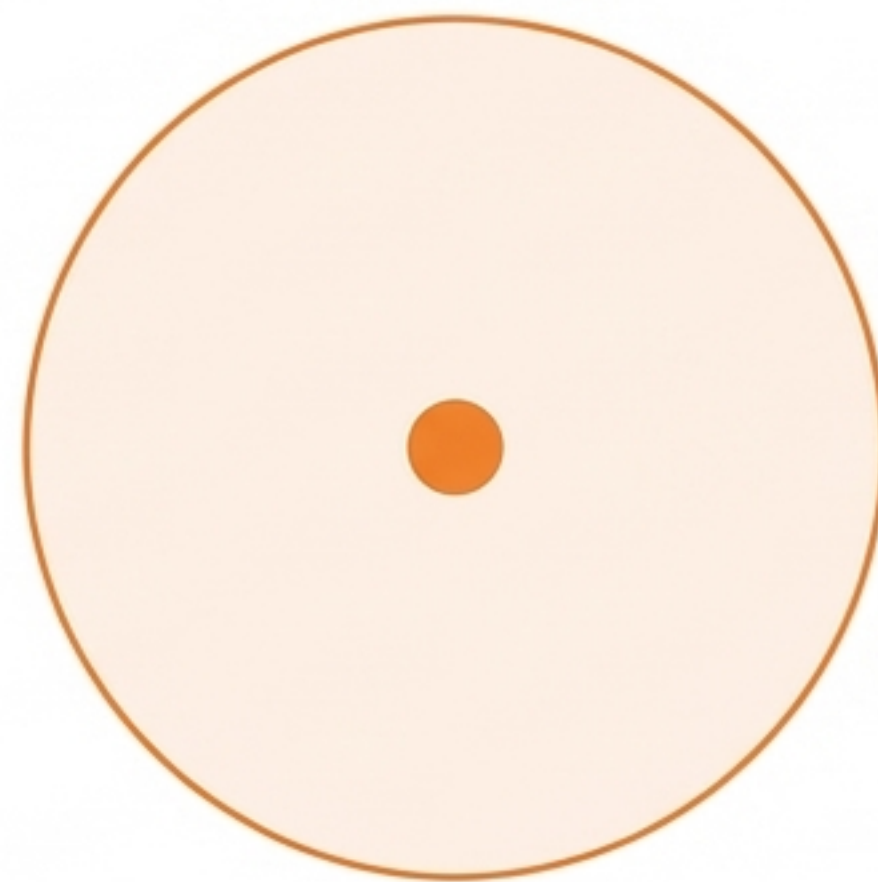
The anatomy of a data space



Core Point:
 ϵ -neighborhood
contains $\geq \text{MinPts}$.

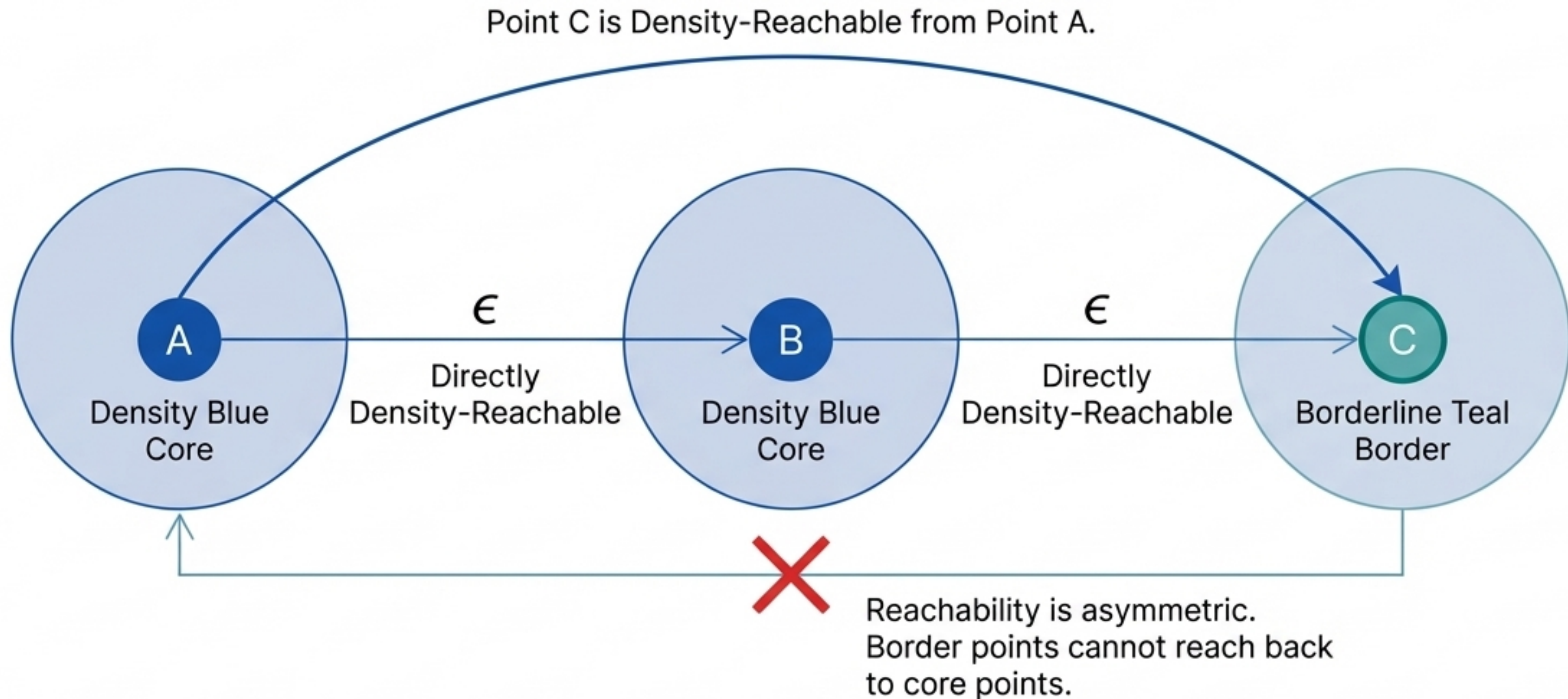


Border Point: $< \text{MinPts}$,
but falls within a Core
point's radius.

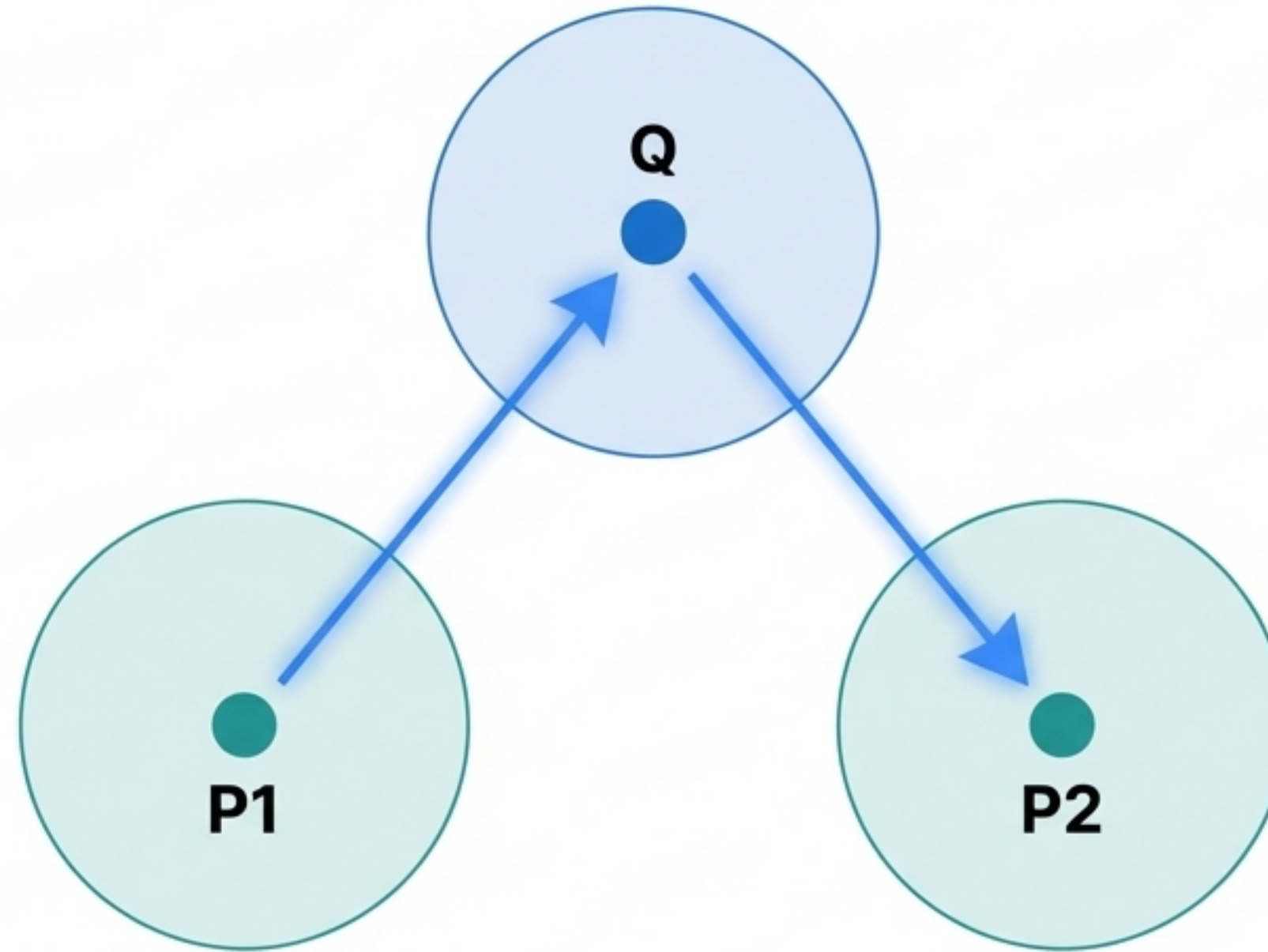


Noise / Outlier:
Isolated and
unreachable.

The directional chain of reachability



Tying the cluster together with density-connectedness

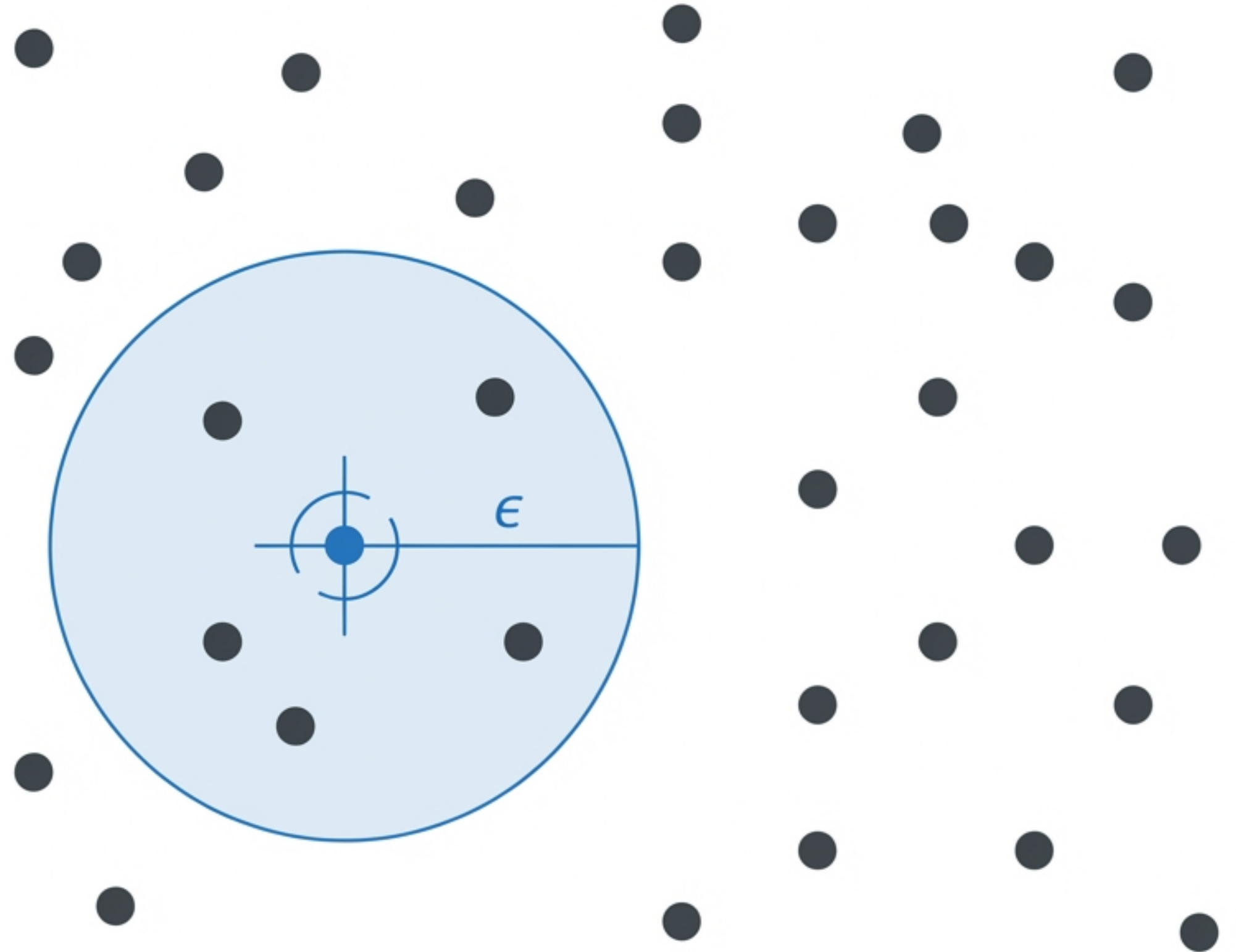


Density-Connectedness is symmetric. P1 and P2 belong to the same cluster because they share a common core ancestor (Q).

Running DBSCAN: The Seed Phase

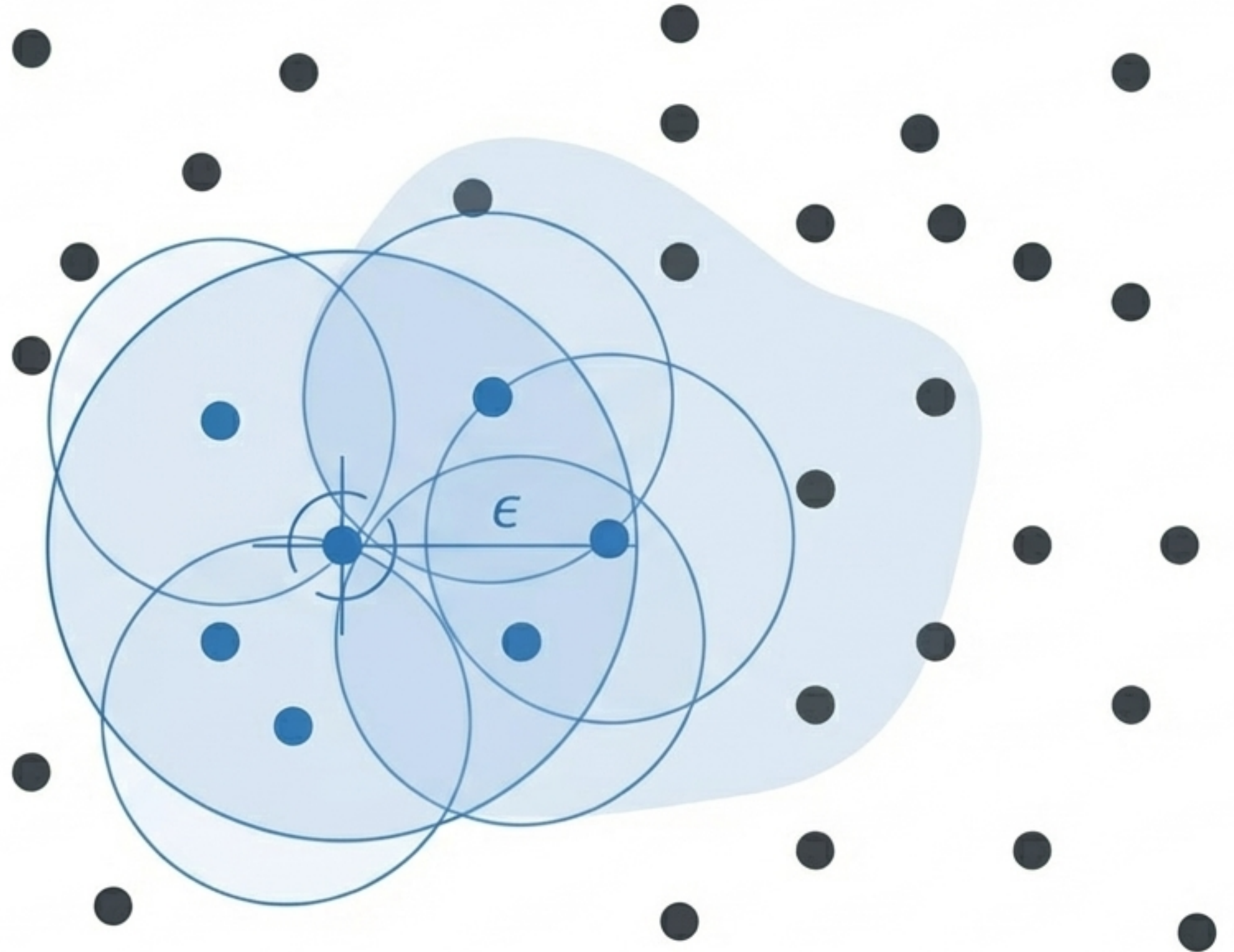
1. Randomly select an unvisited point.

The point passes the MinPts test. It becomes a Core “**Seed**” and a new cluster is born.



Running DBSCAN: The Expansion Phase

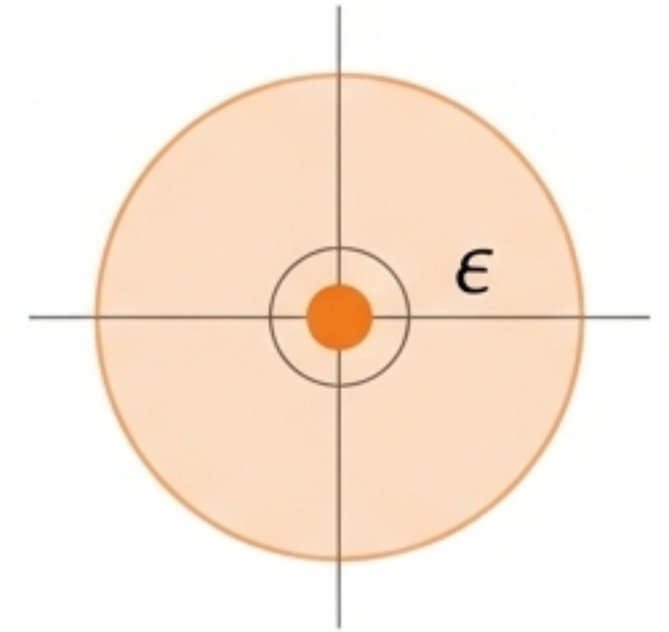
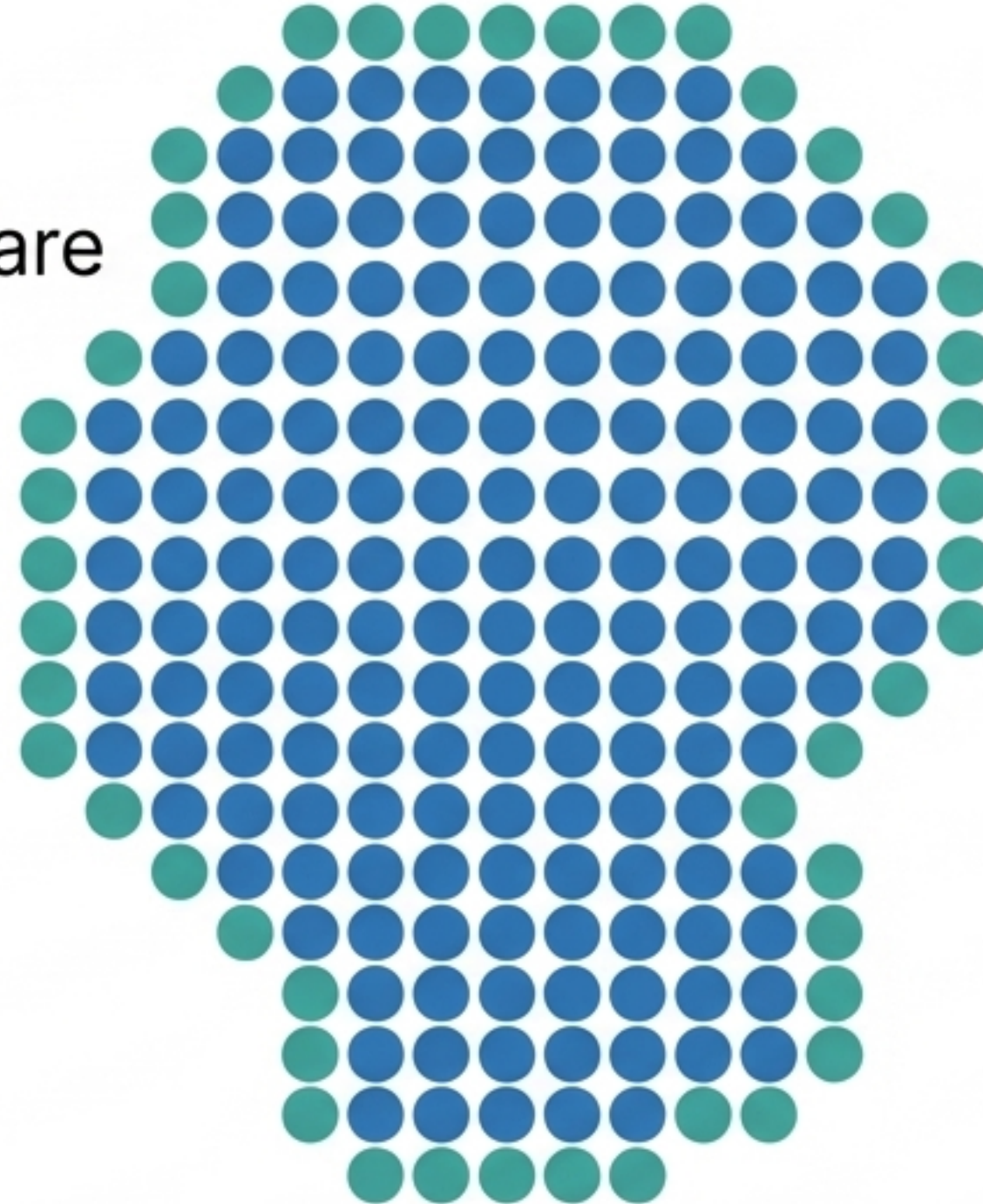
2. Evaluate neighbors.
If they are also Core points, add their neighborhoods to the candidate set. The cluster expands organically.



Running DBSCAN: Termination and Noise

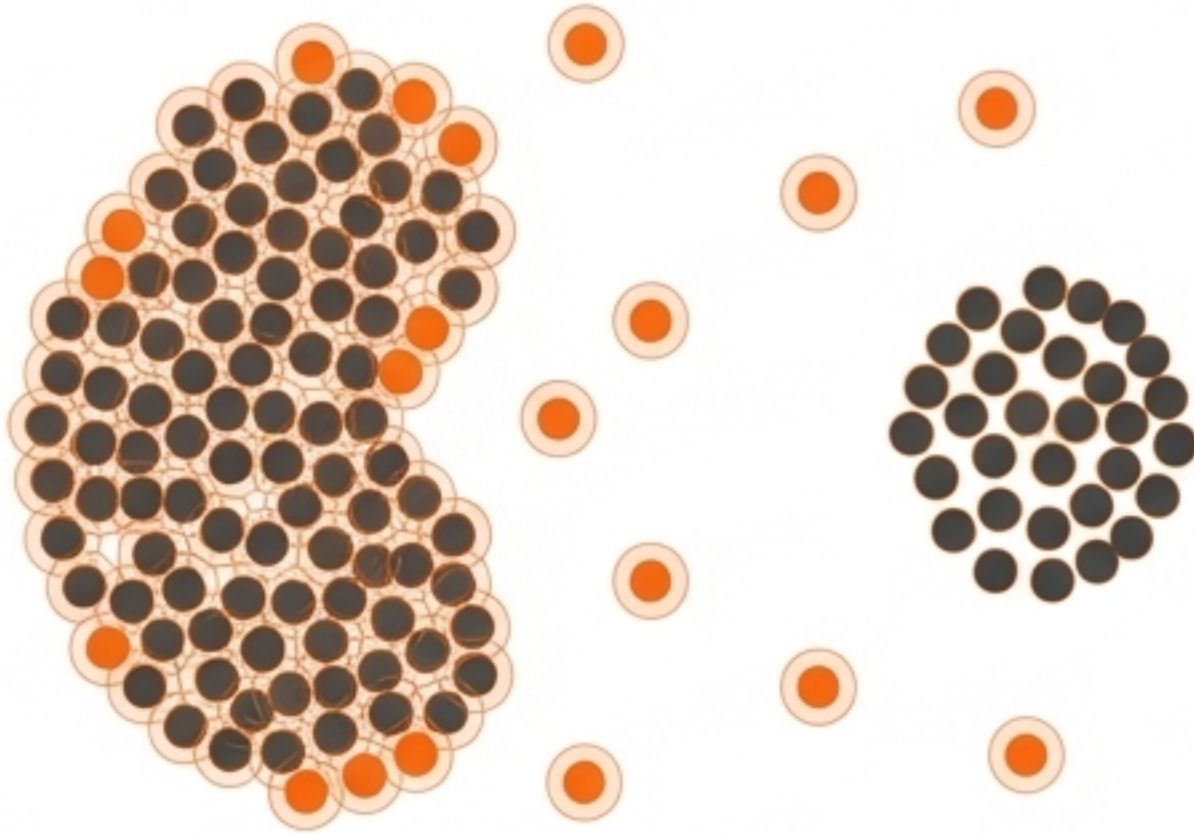
3. Expansion stops at Border points. Unvisited points that fail the density test are flagged as Noise.

Repeat until all points are visited.



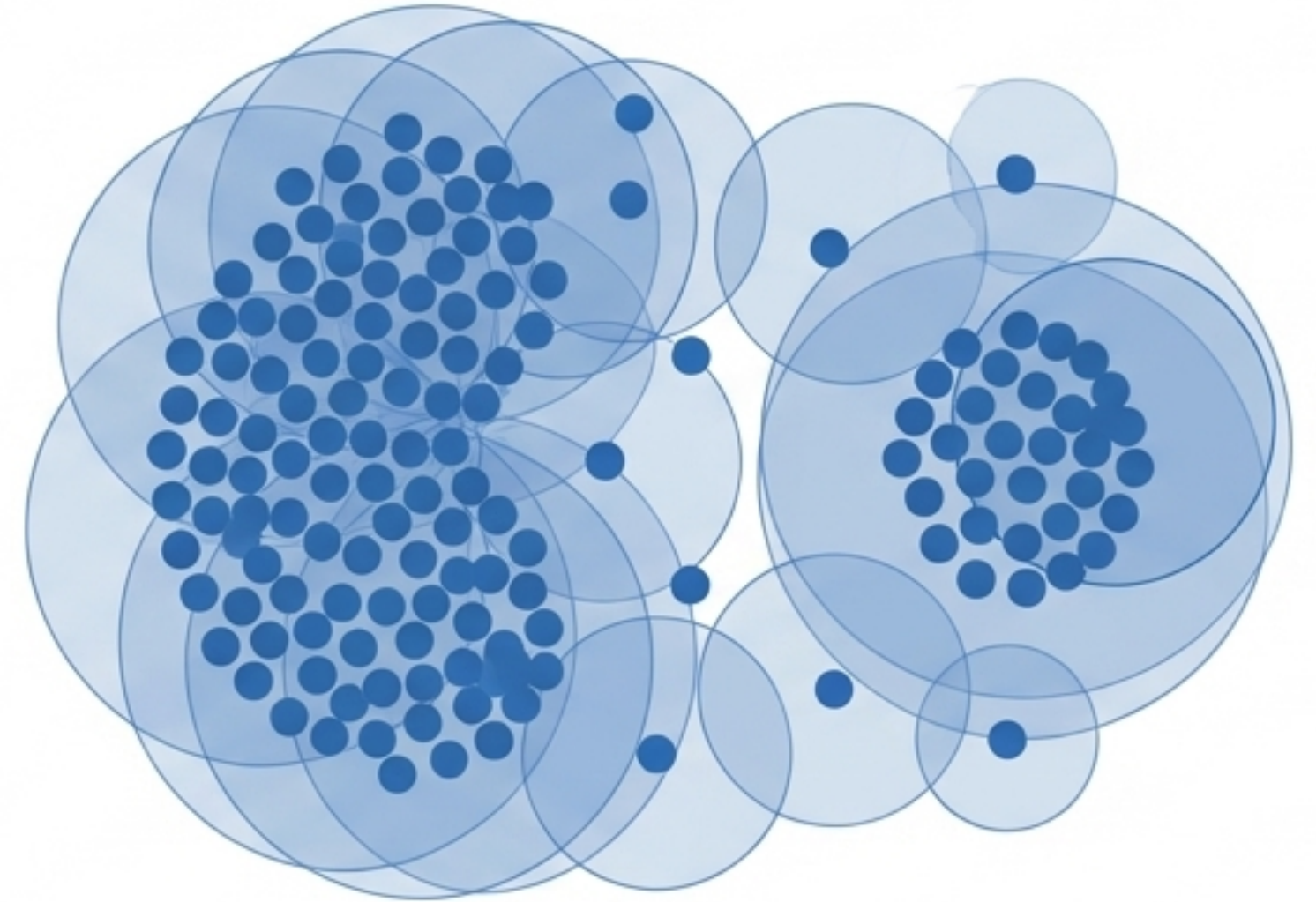
The calibration problem: Finding the perfect radius

ϵ is too small.



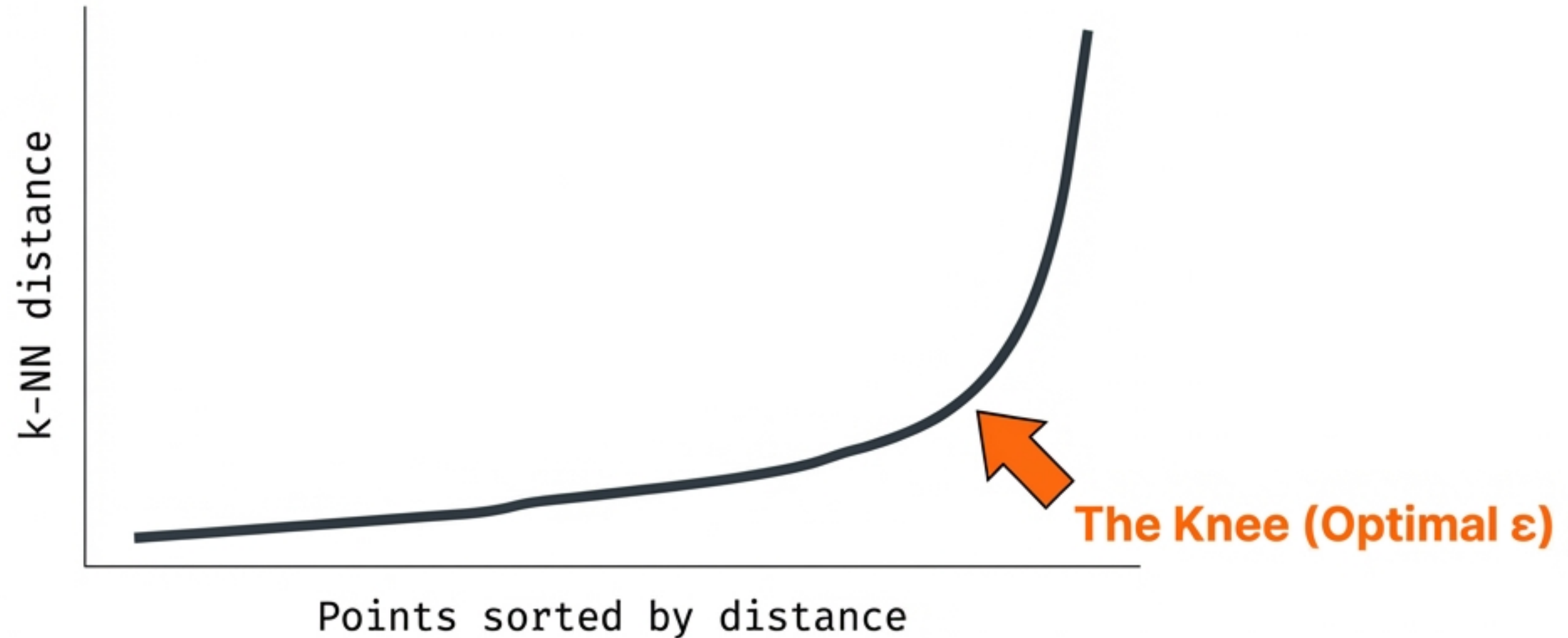
Sparser clusters are incorrectly defined as noise.

ϵ is too large.



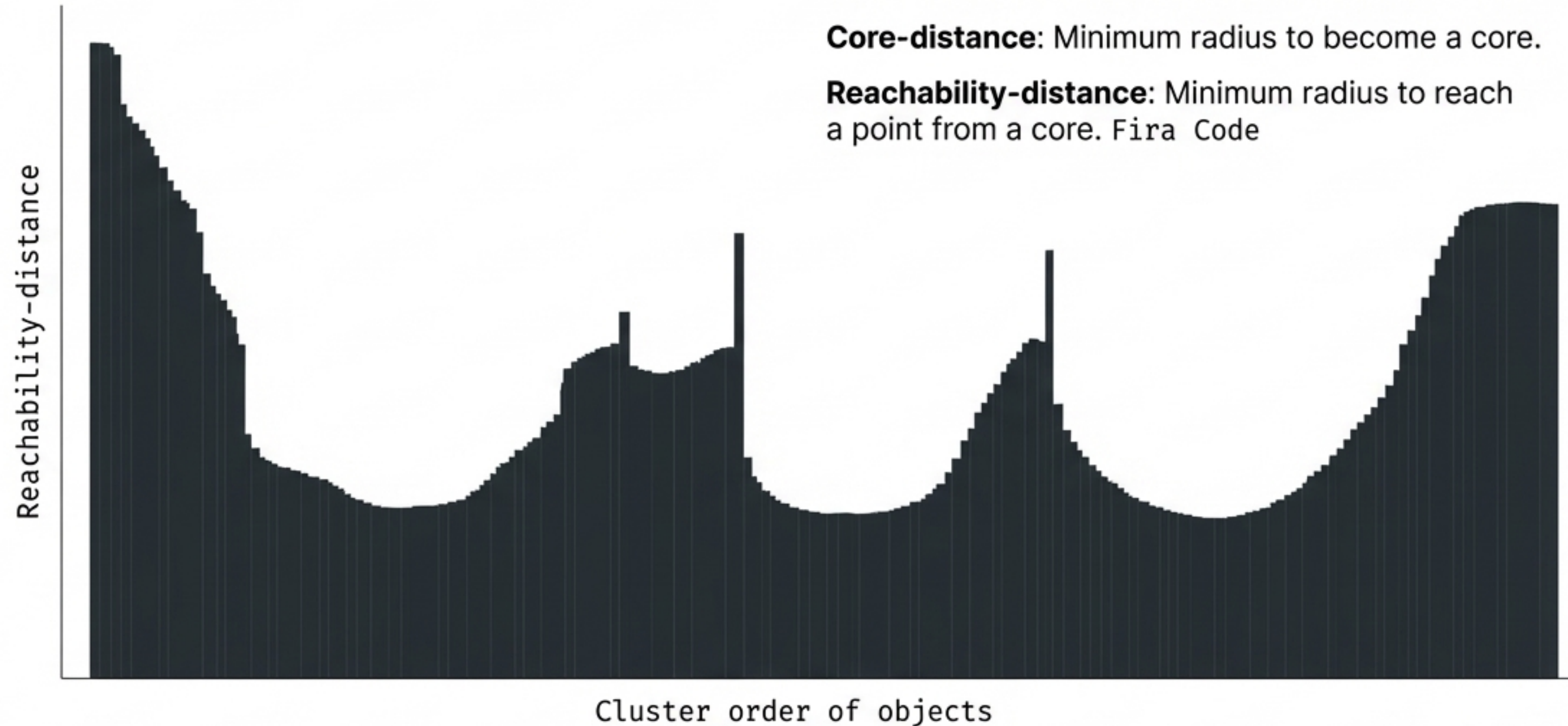
Distinct, dense clusters are incorrectly merged together.

Finding the 'Knee': The k-distance graph



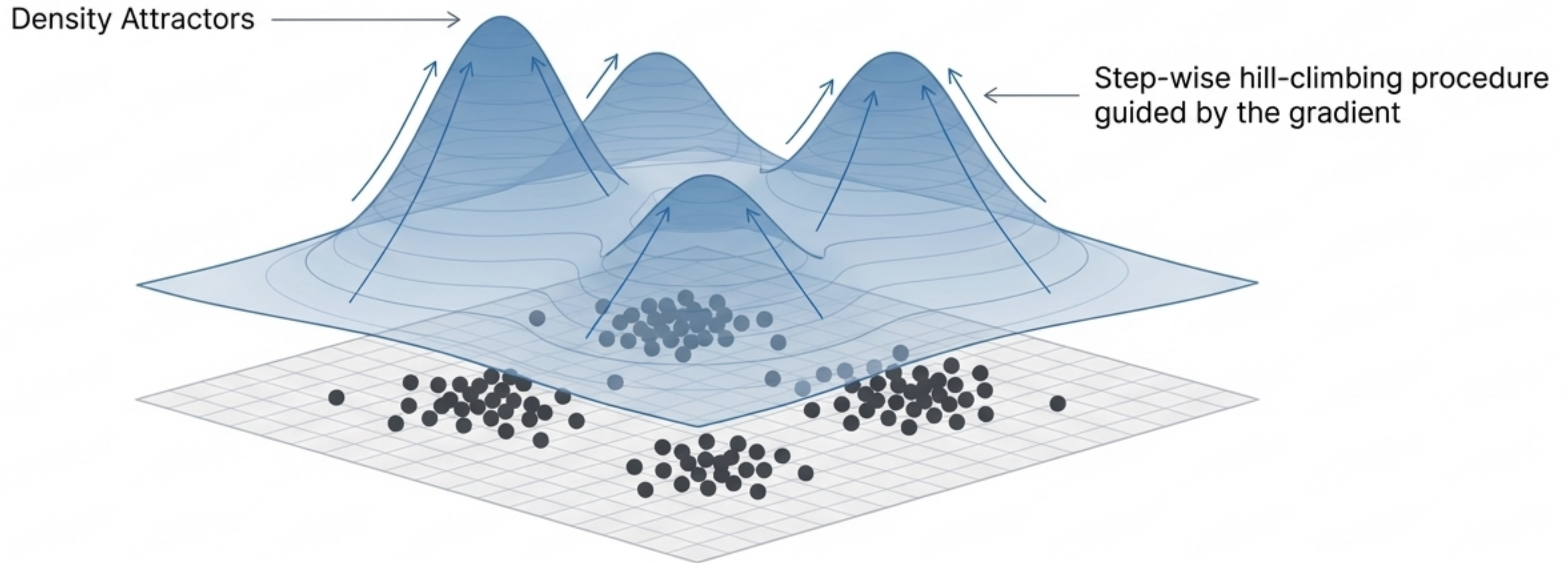
Plot the distance to the k-th nearest neighbor ($k = \text{MinPts}$). The optimal ϵ is located at the threshold where the curve shows a strong upward bend.

Beyond DBSCAN: Handling varying densities with OPTICS



OPTICS does not explicitly produce a hard clustering. Instead, it extracts a hierarchical clustering ordering, allowing you to identify clusters at any density threshold.

Beyond DBSCAN: Continuous distributions with DENCLUE



Instead of radius counting, DENCLUE uses a Gaussian kernel to estimate a continuous density function. Points 'climb' the gradient to assign themselves to a density peak.

The clustering synthesis matrix

	K-Means	DBSCAN	OPTICS	DENCLUE
Arbitrary Shapes?	No (Spherical only)	Yes	Yes	Yes
Handles Noise/Outliers?	No (Highly sensitive)	Yes	Yes	Yes (Invariant against noise)
Primary Input Parameters	k (number of clusters)	ϵ , MinPts	MinPts	Smoothing parameter (h), Noise threshold (ξ)
Handles Varying Densities?	N/A	No (Uses global threshold)	Yes (Outputs cluster ordering)	Yes